4: TRANSLATING ASSEMBLY LANGUAGE INTO MACHINE CODE

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CHAPTER OVERVIEW

4: Translating Assembly Language into Machine Code

Learning Objectives

1. two of the three formats for MIPS instructions.
2. what op codes and functions are for MIPS operations
3. how to format Immediate (I) and Register (R) instructions in machine code.
4. how to represent MIPS instructions in hexadecimal.
5. how to use MARS to check your translations from assembly language to machine code.

Everything in a computer is a binary value. Numbers are binary, characters are binary, and even the instructions to run a program are binary. Therefore the assembly language that has been presented cannot be what the computer understands. This assembly code must be converted to binary values. These binary values are called machine code. This chapter will explain how to convert the assembly instructions that have been covered so far into machine code.

4.1: Instruction Formats
4.2: Machine Code for the Add Instruction
4.3: Machine Code for the Sub Instruction
4.4: Machine Code for the Addi Instruction
4.5: Machine Code for the sll Instruction
4.6: Exercises

4.1: Instruction Formats

The MIPS computer is organized in a 3-address format. Generally all instructions will have 3 addresses associated with them. For example, the instruction " `ADD` R_d , R_s , R_t " uses three registers, the first address is the destination of the result, and the second and third are the two inputs to the ALU. The instruction "ADDI Rt, Rs, immediate" also uses three addresses, but in this case the third address is an immediate value.

In MIPS there are only 3 ways to format instructions. They are the R-format (register), the I- format (immediate), and the J-format (jump). This chapter will only cover R-format and I- format instructions. The R-format and I-format are shown below.

Figure 4-1: R format instruction

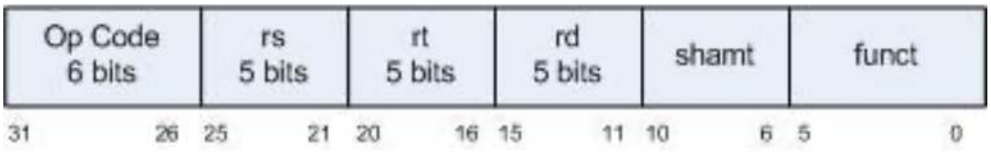
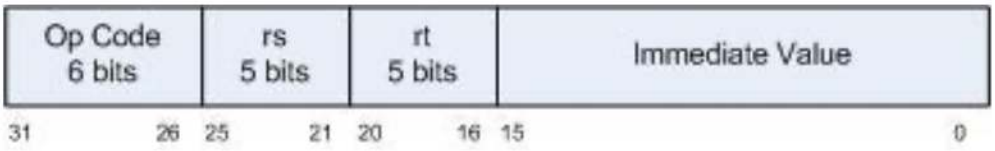


Figure 4-2: I format instruction



The R format instruction is used when the input values to the ALU come from two registers.
The I format instruction is used when the input values to the ALU come from one register and an immediate value.

The fields in these instructions are as follows:

- Op-code - a 6 bit code which specifies the operation. These codes can be found for each instruction on the MIPS Green Sheet found at:

  freepdfs.net/mips-green-sheet...a2139f5cd8d64/.

  Op-codes these are stored as a 2 digit number. The first number is 2 bits wide, and has values from 0-2. The second number is 4 bits wide, and has values from 0x0 to 0xf. So for example the `lw` instruction has an op code of 23, or 100011_2. The `addi` has an op code of 8, or 001000 (if the first digit is not specified, it is zero).

  All R-format operators have an op-code of 0, and are specified as an op-code/function. So the `add` op-code is specified by an op-code/function of 0/20. The operation to be performed by the ALU is contained in the function, which is the last field in the R instruction. Functions, like op-codes, are specified as a 2 bit number and a 4 bit number.

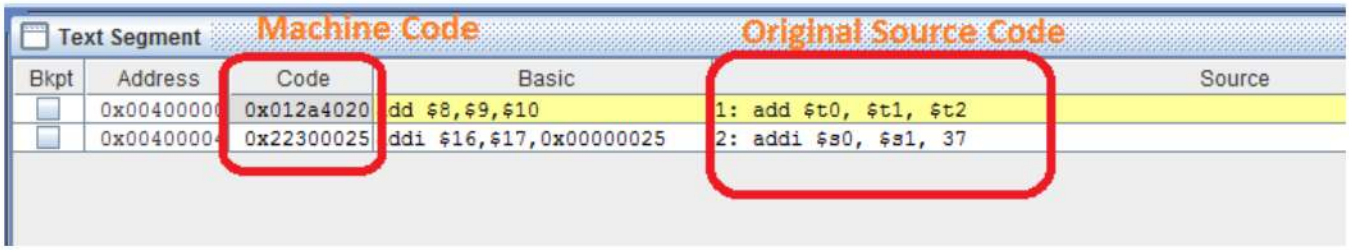
- rs - This is the first register in the instruction, and is always an input to the ALU. Note that rs, rt, and rd are 5 bits wide, which allows the addressing of the 32 general purpose registers.
- rt - This is the second register in the instruction. It is the second input to the ALU when there is an rd specified in the instruction (as in ADD) and the destination register when rd is not specified in the instruction (as in ADDI)
- rd - rd always is the destination register when it is specified.
- shamt - The number of bits to shift if the operation is a shift operation, otherwise it is 0.

  It is 5 bits wide, which allows for shifting of all 32 bits in the register.

- funct - For an R type instruction, this specifies the operation for the ALU.
- immediate - The immediate value for instructions which use immediate values.

In MARS the machine code which is generated from the assembly code is shown in the Code column, as in figure 4-3. A good way to check if you have translated your code correctly is to see if your translation matches the code in this column.

Figure 4-3: Machine Code example



The rest of the sections of this chapter are intended to help makes sense of this machine code.

4.2: Machine Code for the Add Instruction

This section will translate the following add instruction to machine code.

```
add $t0, $t1, $t2
```

The MIPS Greensheet specifies the add instruction as an R-format instruction and the op- code/function for the add as 0/20. The op-code/function field is made up of two numbers, the first is the op-code, and the second is the function. Note that the function is used only for R format instructions. If the instruction has a function, the number to the left of the "/" is the op- code, and the number to the right of the "/" is the function. If there is only one number with no "/", it is the op-code, and there is no function.

Both the op-code and the function are 6 bits, divided into a 2 bit number and a 4 bit number. So the first number in the op-code and function is 0..3, and the second is 0..f. Both are generally called hex values, so this text will do so as well. So the 6 bits for the op-code translate to 00 0000, and the 6 bits for the function translate to 10 0000. These are placed into the op-code and function fields of the R format instruction shown in figure 4-5 below.

Register Rd is `$t0` . `$t0` is also register `$8` , or 01000, so 01000 is placed in the R_d field.

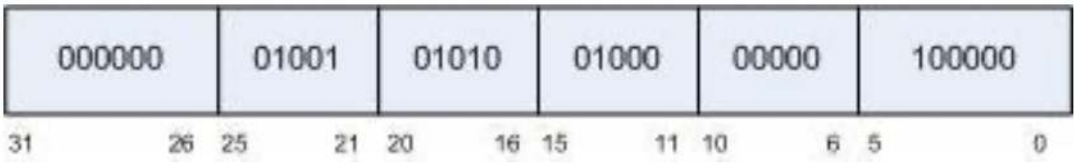
Register Rs is `$t1` . `$t1` is also register `$9` , or 01001, so 01001 is placed in the R_s field

Register Rt is `$t2` . `$t2` is also register `$10` , or 01010, so 01010 is placed in the R_t field.

The shamt is 00000 as there are no bits being shifted.

The result is the following R-format instruction.

Figure 4-4: Machine code for add $t0, $t1, $t2



Thus the instruction " `add $t0 , $t1 , $t2` " translate into the bit string

00000001001010100100000000100000

This is as hard to type as it is to read. To make it readable, the bits are divided into groups of 4 bits, and these 4 bit values translated into hex. This results in the following instruction.

	0000	0001	0010	1010	0100	0000	0010	0000
0x	0	1	2	a	4	0	2	0

This can be checked by entering the instruction in MARS, and assembling it to see the resulting machine code. Be careful with the order of the registers when translating into machine code, as the Rd is the last register in machine code.

4.3: Machine Code for the Sub Instruction

This section will translate the following sub instruction to machine code.

```
sub $s0, $s1, $s2
```

The MIPS Greensheet specifies the `sub` instruction as an R-format instruction and the op- code/function for the `sub` as 0/22. This means the 6 bits for the op code are 000000 and the 6 bits for the function are 100010.

Register R_d is `$s0` is also register `$16` , or `10000` .

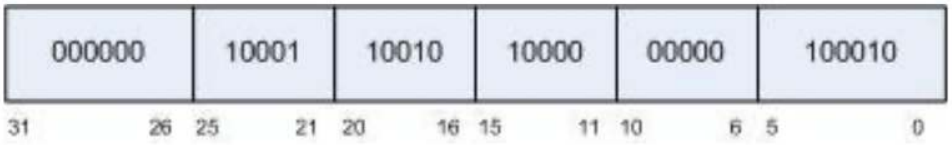
Register R_s is `$s1` is also register `$17` , or `10001` .

Register R_t is `$s2` is also register `$18` , or `10010` .

The shamt is `00000` as there are no bits being shifted.

The result is the following R-format instruction.

Figure 4-5: Machine code for sub $s0, $s1, $s2



To write this instruction's machine code, the bits are organized in groups of 4, and hex values given. This results in the number 0000 0010 0011 0010 1000 0000 0010 00102 or 0x02328022.

4.4: Machine Code for the Addi Instruction

This section will translate the following addi instruction to machine code.

```
addi $s2, $t8, 37
```

The MIPS Greensheet specifies the `addi` instruction as an I-format instruction and the op- code/function for the `addi` as 8 (note that there is no function for an I-format instruction). This means the 6 bits for the op code are 001000.

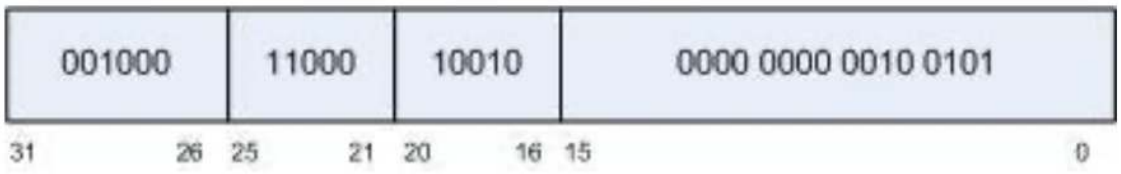
Register R_t is `$s2` is also register `$18` , or `10010` .

Register R_s is `$t8` is also register `$24` , or `11000` .

The immediate value is 37, or `0x0025` .

The Result is the following I-format instruction.

Figure 4-6: Machine code for addi $s2, $t8, 37



To write this instruction's machine code, the bits are organized in groups of 4, and hex values given. This results in the number 0010 0011 0001 0010 0000 0000 0010 01012 or 0x23120025.

When translating the I-format instruction, be careful to remember that R_t is destination register.

4.5: Machine Code for the sll Instruction

This section will translate the following SLL instruction to machine code.

```
sll $t0, $t1, 10
```

The MIPS Greensheet specifies the `sll` instruction as an R-format instruction and the op- code/function for the `sll` as 0/00. This means the 6 bits for the op code are 000000 and the 6 bits for the function are 000000.

Register rd is `$t0` is also register `$8` , or `01000` .

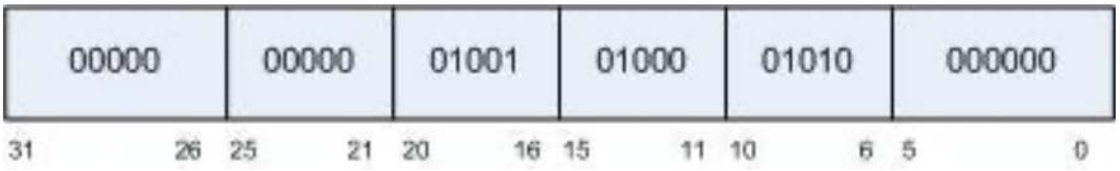
Register rs is not used.

Register rt is `$t1` is also register `$9` , or `01001` .

The shamt is 10, or `0x01010` .

The Result is the following R-format instruction.

Figure 4-7: Machine code for sll $t0, $t1, 10



4.6: Exercises

1. Translate the following MIPS assembly language program into machine code.

```
.text
.globl main
main:
    ori $t0, $zero, 15
    ori $t1, $zero, 3
    add $t1, $zero $t1
    sub $t2, $t0, $t1
    sra $t2, $t2, 2
    mult $t0, $t1
    mflo $a0
    ori $v0, $zero, 1
    syscall
    addi $v0, $zero, 10
    syscall

.data
result: .asciiz "15 * 3 is "
```

2. Translate the following MIPS machine code into MIPS assembly language.

```
0x2010000a
0x34110005
0x012ac022
0x00184082
0x030f9024
```